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Inventor(s)	Mickelsen; Steven R.

Methods and apparatus for selective tissue ablation

Abstract

Catheter systems and methods for the selective and rapid application of DC voltage to drive irreversible electroporation are disclosed herein. In some embodiments, an apparatus includes a voltage pulse generator and an electrode controller. The voltage pulse generator is configured to produce a pulsed voltage waveform. The electrode controller is configured to be operably coupled to the voltage pulse generator and a medical device including a series of electrodes. The electrode controller includes a selection module and a pulse delivery module. The selection module is configured to select a subset of electrodes from the series of electrodes. The selection module is configured to identify at least one electrode as an anode and at least one electrode as a cathode. The pulse delivery module is configured to deliver an output signal associated with the pulsed voltage waveform to the subset of electrodes.

Inventors:	Mickelsen; Steven R. (Iowa City, IA)
Applicant:	Boston Scientific Scimed, Inc. (Maple Grove, MN)
Family ID:	53277039
Assignee:	Boston Scientific Scimed, Inc. (Maple Grove, MN)
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Primary Examiner: Vahdat; Khadijeh A

Attorney, Agent or Firm: Nelson Mullins Riley & Scarborough LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/349,299, filed Jun. 16, 2021, which is a continuation of U.S. patent application Ser. No. 15/795,062, filed Oct. 26, 2017, now U.S. Pat. No. 11,259,869, which is a continuation of U.S. patent application Ser. No. 15/341,512, filed Nov. 2, 2016, now Abandoned, which is a continuation of PCT Application No. PCT/US2015/029734, filed May 7, 2015, which claims priority to U.S. Provisional Application Ser. No. 61/996,390, filed May 7, 2014, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

(1) The embodiments described herein relate generally to medical devices for therapeutic electrical energy delivery, and more particularly to systems and methods for delivering electrical energy in the context of ablating tissue rapidly and selectively by the application of suitably timed pulsed voltages that generate irreversible electroporation of cell membranes.

(2) The past two decades have seen advances in the technique of electroporation as it has progressed from the laboratory to clinical applications. Known methods include applying brief, high voltage DC pulses to tissue, thereby generating locally high electric fields, typically in the range of hundreds of Volts/centimeter. The electric fields disrupt cell membranes by generating

pores in the cell membrane, which subsequently destroys the cell membrane and the cell. While the precise mechanism of this electrically-driven pore generation (or electroporation) awaits a detailed understanding, it is thought that the application of relatively large electric fields generates instabilities in the phospholipid bilayers in cell membranes, as well as mitochondria, causing the occurrence of a distribution of local gaps or pores in the membrane. If the applied electric field at the membrane exceeds a threshold value, typically dependent on cell size, the electroporation is irreversible and the pores remain open, permitting exchange of material across the membrane and leading to apoptosis or cell death. Subsequently, the surrounding tissue heals in a natural process. (3) While pulsed DC voltages are known to drive electroporation under the right circumstances, the examples of electroporation applications in medicine and delivery methods described in the prior art do not discuss specificity of how electrodes are selected to accomplish the desired ablation. (4) There is a need for selective energy delivery for electroporation and its modulation in various tissue types, as well as pulses that permit rapid action and completion of therapy delivery. There is also a need for more effective generation of voltage pulses and control methods, as well as appropriate devices or tools addressing a variety of specific clinical applications. Such more selective and effective electroporation delivery methods can broaden the areas of clinical application of electroporation including therapeutic treatment of a variety of cardiac arrhythmias.

SUMMARY

(5) Catheter systems and methods are disclosed for the selective and rapid application of DC voltage to drive electroporation. In some embodiments, an apparatus includes a voltage pulse generator and an electrode controller. The voltage pulse generator is configured to produce a pulsed voltage waveform. The electrode controller is configured to be operably coupled to the voltage pulse generator and a medical device including a series of electrodes. The electrode controller includes a selection module and a pulse delivery module. The selection module is configured to select a subset of electrodes from the series of electrodes. The selection module is configured to identify at least one electrode as an anode and at least one electrode as a cathode. The pulse delivery module is configured to deliver an output signal associated with the pulsed voltage waveform to the subset of electrodes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a portion of a catheter with multiple electrodes disposed along its shaft and wrapped around the pulmonary veins of the heart such that the portion of the catheter is placed in the epicardial space of the heart.
- (2) FIG. 2A is a schematic illustration of an irreversible electroporation system that includes a DC voltage/signal generator, a controller capable of being configured to apply voltages to selected subsets of electrodes with independent subset selections for anode and cathode.
- (3) FIG. 2B is a schematic illustration of an irreversible electroporation system that includes a DC voltage/signal generator and a controller according to an embodiment.
- (4) FIG. 3 is an illustration of an ECG waveform showing the refractory periods during atrial and ventricular pacing, and the time windows for irreversible electroporation ablation.
- (5) FIG. 4 is a schematic illustration of a user interface according to an embodiment showing various configurations for subsets of electrodes on the catheter to be identified as anodes or cathodes, and the ablation vectors therebetween.
- (6) FIG. 5 is a schematic illustration of a user interface according to an embodiment, indicating the electrodes that are active and wrapped around the pulmonary veins of a patient anatomy, and that can be actively available for selection in bipolar ablation.
- (7) FIG. 6 is a schematic illustration of a user interface according to an embodiment that illustrates

a method of ablating along a cardiac isthmus line using two catheters, one of which is wrapped around the pulmonary veins.

(8) FIG. 7 is a schematic illustration of a user interface according to an embodiment that illustrates a method of focal ablation, wherein nearby electrodes on the same catheter are used for bipolar ablation with irreversible electroporation to ablate a region of contacting tissue at the distal catheter end, and showing a depth selection button that can be used to selectively adjust an ablation parameter.

(9) FIG. 8 is a schematic illustration according to an embodiment, showing a graphic window for selection of type of pacing to be used for electroporation therapy delivery.

(10) FIG. 9 is a schematic illustration of a user interface according to an embodiment, showing a graphic window with parameters associated with a selected pacing type.

(11) FIG. 10 is a schematic illustration of a user interface that illustrates a method of selecting anode electrode subsets and cathode electrode subsets on two different catheters according to an embodiment.

(12) FIG. 11 is a schematic illustration of a user interface that illustrates a method of selecting a single anode electrode and a single selected cathode electrode on the same catheter according to an embodiment.

(13) FIG. 12 is a schematic illustration of a waveform generated by the irreversible electroporation system and methods according to an embodiment, showing a balanced square wave.

(14) FIG. 13 is a schematic illustration of a waveform generated by the irreversible electroporation system and methods according to an embodiment, showing a balanced biphasic square wave.

(15) FIG. 14 is a schematic illustration of a waveform generated by the irreversible electroporation system and methods according to an embodiment, showing a progressive balanced biphasic square wave.

(16) FIG. 15 is a schematic illustration of a signal generator according to an embodiment, showing a capacitor bank connected to a set of switches.

(17) FIG. 16 is a schematic illustration of a voltage pulse-time plot produced by a high voltage DC signal generator according to an embodiment.

(18) FIG. 17 is a schematic depiction of a sequence of three biphasic pulses delivered by a voltage generator and using methods according to an embodiment, showing a pre-polarizing negative voltage pulse followed by a positive polarizing pulse.

(19) FIG. 18A is a schematic illustration of a distal portion of a catheter having two electrodes functioning as a cathode and an anode according to an embodiment.

(20) FIG. 18B is a schematic illustration of a distal portion of a catheter according to an embodiment having a recessed distal anode electrode, the illustration showing the current flow between electrodes.

(21) FIG. 18C is a schematic illustration of a distal portion of a catheter according to an embodiment, where electric field lines near the distal end of the catheter are schematically shown together with a region of focal ablation in tissue.

(22) FIG. 18D is a schematic illustration of a distal portion of a catheter according to an embodiment, where the catheter lumen has flowing fluid, possibly with ultrasonic waves conducted through the fluid, and the distal anode electrode has channels for fluid flow to exit the distal end of the catheter.

(23) FIG. 19A is an illustration of the simulation geometry showing a geometric model of the distal portion of a catheter and surrounding tissue types.

(24) FIG. 19B is an illustration of the distal portion of a catheter with a proximal electrode disposed externally on the shaft of the catheter and a distal electrode that is recessed and disposed on the inner side of the shaft of the catheter.

(25) FIG. 19C is an illustration of a longitudinal cross-section of the simulation geometry showing distal catheter geometry and tissue geometry together with electric field intensity contours for the

case when the proximal electrode is disposed externally on the shaft of the catheter and the distal electrode is recessed and disposed on the inner side of the shaft of the catheter, according to an embodiment.

(26) FIG. 19D is an illustration of a longitudinal cross-section of the simulation geometry showing distal catheter geometry and tissue geometry together with electric field intensity contours for the case when the proximal and distal electrodes are both disposed externally on the shaft of the catheter, according to an embodiment.

(27) FIG. 20 is a schematic illustration of a balloon catheter according to an embodiment, where a balloon structure with an insulating exterior is located between anode and cathode electrodes for electroporation voltage delivery.

(28) FIG. 21A is a schematic illustration of a balloon catheter according to an embodiment, where a balloon structure with an insulating exterior and with at least one ultrasonic transducer is located between anode and cathode electrodes for electroporation voltage delivery.

(29) FIG. 21B is a schematic illustration of a balloon catheter according to an embodiment, where an annular ablation region abutting a vessel is shown.

(30) FIG. 22 is a schematic illustration of a basket ablation catheter for electroporation energy delivery according to an embodiment.

DETAILED DESCRIPTION

(31) Systems and methods are disclosed for the selective and rapid application of DC voltage to drive electroporation. In some embodiments, the irreversible electroporation system described herein includes a DC voltage/signal generator and a controller capable of being configured to apply voltages to a selected multiplicity or a subset of electrodes, with anode and cathode subsets being selected independently. The controller is additionally capable of applying control inputs whereby selected pairs of anode-cathode subsets of electrodes can be sequentially updated based on a pre-determined sequence.

(32) In some embodiments, an irreversible electroporation system includes a DC voltage/signal generator and a controller capable of being configured to apply voltages to a selected multiplicity or a subset of electrodes, with independent subset selections for anode and cathode. Further, the controller is capable of applying control inputs whereby selected pairs of anode-cathode subsets of electrodes can be sequentially updated based on a predetermined sequence. The generator can output waveforms that can be selected to generate a sequence of voltage pulses in either monophasic or biphasic forms and with either constant or progressively changing amplitudes. In one embodiment, a DC voltage pulse generation mechanism is disclosed that can use amplified voltage spikes generated by the action of a switch connected to a capacitor bank, resulting in a biphasic, asymmetric micro pulse waveform for cardiac ablation where the first phase provides a brief pre-polarizing pulse that is followed by a finishing pulse in the second phase. Such pre-polarization can result in more effective pulsed voltage electroporation delivery. Methods of control and DC voltage application from a generator capable of selective excitation of sets of electrodes are disclosed. Devices are disclosed for more effective DC voltage application through ionic fluid irrigation and ultrasonic agitation, possibly including insulating balloon constructions to displace electrodes from collateral structures, and for use in intravascular applications. Such devices for generating irreversible electroporation can be utilized in cardiac therapy applications such as ablation to treat Ventricular Tachycardia (VT) as well as in intravascular applications. In some embodiments, the use of temperature to selectively ablate tissue is described, as the threshold of irreversible electroporation is temperature-dependent, utilizing focused kinetic energy to select the predominant tissue type or region it is desired to ablate.

(33) In some embodiments, an apparatus includes a voltage pulse generator and an electrode controller. The voltage pulse generator is configured to produce a pulsed voltage waveform. The electrode controller is configured to be operably coupled to the voltage pulse generator and a medical device including a series of electrodes. The electrode controller is implemented in at least

one of a memory or a processor, and includes a selection module and a pulse delivery module. The selection module is configured to select a subset of electrodes from the series of electrodes. The selection module is configured identify at least one electrode as an anode and at least one electrode as a cathode. The pulse delivery module is configured to deliver an output signal associated with the pulsed voltage waveform to the subset of electrodes.

(34) In some embodiments, an apparatus includes a voltage pulse generator and an electrode controller. The voltage pulse generator is configured to produce a pulsed voltage waveform. The electrode controller is configured to be operably coupled to the voltage pulse generator and a medical device including a series of electrodes. The electrode controller is implemented in at least one of a memory or a processor, and includes a selection module and a pulse delivery module. The selection module is configured to select a set of anode/cathode pairs, each anode/cathode pair including at least one anode electrode and at least one cathode electrode. In some embodiments, for example, the anode/cathode pair can include one anode and multiple cathodes (or vice-versa). The pulse delivery module is configured to deliver an output signal associated with the pulsed voltage waveform to the plurality of anode/cathode pairs according to a sequential pattern.

(35) In some embodiments, a non-transitory processor readable medium storing code representing instructions to be executed by a processor includes code to cause the processor to identify a set of anode/cathode pairs from a set of electrodes of a multi-electrode catheter. The multi-electrode catheter is configured to be disposed about a portion of a heart, and at least one anode/cathode pair includes at least one anode electrode and at least one cathode electrode. The code further includes code to convey a pacing signal to a pacing lead configured to be operatively coupled to the heart. The code further includes code to receive an electrocardiogram associated with a function of the heart. The code further includes code to deliver a pulsed voltage waveform to the set of anode/cathode pairs according to a sequential pattern.

(36) In some embodiments, a method includes identifying, via a selection module of an electrode controller, a set of anode/cathode pairs from a set of electrodes of a multi-electrode catheter. The multi-electrode catheter is configured to be disposed about a portion of a heart. At least one anode/cathode pair includes at least one anode electrode and at least one cathode electrode. A pacing signal is conveyed to a pacing lead configured to be operatively coupled to the heart. The method further includes receiving, at a feedback module of the electrode controller, an electrocardiogram associated with a function of the heart. The method further includes delivering, via a pulse delivery module of the electrode controller, a pulsed voltage waveform to the set of anode/cathode pairs according to a sequential pattern.

(37) In some embodiments, an apparatus includes a signal generator for the generation of DC voltage pulses. The signal generator is configured to produce a biphasic waveform having a pre-polarizing pulse followed by a polarizing pulse. The pre-polarizing pulse is generated by utilizing voltage spikes generated from switching on a discharge of a capacitor bank.

(38) In some embodiments, an apparatus includes a catheter shaft, a cathode electrode and an anode electrode. The catheter shaft has an outer side and an inner side. The cathode electrode is coupled to a distal end portion of the catheter shaft such that a cathode surface is exposed on the outer side of the catheter shaft. The anode electrode is coupled to the distal end portion distal relative to the cathode electrode. The anode electrode is recessed within the catheter shaft and coupled to the catheter shaft such that an anode surface is exposed on the inner side of the catheter shaft.

(39) In some embodiments, an apparatus includes a catheter shaft, an inflatable balloon, a first electrode and a second electrode. The catheter shaft has a distal end portion. The inflatable balloon is coupled to the distal end portion. An outer surface of the balloon is an electrical insulator. The first electrode is coupled to a proximal side of the balloon, and the second electrode is coupled to a distal side of the balloon.

(40) In some embodiments, an apparatus includes a catheter shaft having a distal end portion, an expandable basket structure, a first electrode, a second electrode, and a set of spherical electrodes.

The expandable basket structure is coupled to the distal end portion of the catheter shaft. The first electrode coupled to a proximal side of the expandable basket structure. The second electrode is coupled to a distal side of the expandable basket structure. The set of spherical electrodes is coupled to a corresponding set of rounded corners of the expandable basket structure.

(41) As used in this specification, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, the term “a member” is intended to mean a single member or a combination of members, “a material” is intended to mean one or more materials, “a processor” is intended to mean a single processor or multiple processors; and “memory” is intended to mean one or more memories, or a combination thereof.

(42) As used herein, the terms “about” and “approximately” generally mean plus or minus 10% of the value stated. For example, about 0.5 would include 0.45 and 0.55, about 10 would include 9 to 11, about 1000 would include 900 to 1100.

(43) As shown in FIG. 1, in some embodiments a Pulmonary Vein isolation (PV isolation) ablation catheter device **15** with a multiplicity of electrodes (indicated by dark bands such as those marked as **17**) disposed along its mid-section is wrapped in the epicardial space around the pulmonary veins **10**, **11**, **12** and **13** of a heart **7** in a subject or patient anatomy, with the ends **8** and **9** of the mid-section extending out and away to eventually emerge from the patient's chest. The ablation catheter **15** and any of the catheters described herein can be similar to the ablation catheters described in PCT Publication No. WO2014/025394, entitled “Catheters, Catheter Systems, and Methods for Puncturing Through a Tissue Structure,” filed on Mar. 14, 2013 (“the '394 PCT Application”), which is incorporated herein by reference in its entirety. The ablation catheter **15** can be disposed about the pulmonary veins **10**, **11**, **12** and **13** using any suitable procedure and apparatus. For example, in some embodiments, the ablation catheter **15** can be disposed about the pulmonary veins **10**, **11**, **12** and **13** and/or the heart **7** using a puncturing apparatus disposed via a subxiphoid pericardial access location and a guidewire-based delivery method as described in the '394 PCT Application. After the ends **8** and **9** of the mid-section extend and emerge out of the patient chest they can be cinched together to effectively hold the catheter in place or position.

(44) A DC voltage for electroporation can be applied to subsets of electrodes identified as anode and cathode, respectively, on approximately opposite sides of the closed contour defined by the shape of the catheter **15** around the pulmonary veins. The DC voltage is applied in brief pulses sufficient to cause irreversible electroporation. In any of the systems and methods described herein, the pulse or waveform can be in the range of 0.5 kV to 10 kV and more preferably in the range 1 kV to 2.5 kV, so that a threshold electric field value of around 200 Volts/cm is effectively achieved in the cardiac tissue to be ablated. In some embodiments, the marked electrodes can be automatically identified, or manually identified by suitable marking, on an X-ray or fluoroscopic image obtained at an appropriate angulation that permits identification of the geometric distance between anode and cathode electrodes, or their respective centroids. In one embodiment, the DC voltage generator setting for irreversible electroporation is then automatically identified by the electroporation system based on this distance measure. In an alternate embodiment, the DC voltage value is selected directly by a user from a suitable dial, slider, touch screen, or any other user interface. The DC voltage pulse results in a current flowing between the anode and cathode electrodes on opposite sides of the contour defined by the catheter shape, with said current flowing through the cardiac wall tissue and through the intervening blood in the cardiac chamber, with the current entering the cardiac tissue from the anode and returning back through the cathode electrodes. The forward and return current paths (leads) are both inside the catheter. Areas of cardiac wall tissue where the electric field is sufficiently large for irreversible electroporation are ablated during the DC voltage pulse application. The number of electrodes on the PV isolation ablation catheter can be in the range between 8 and 50, and more preferably in the range between 15 and 40.

(45) A schematic diagram of the electroporation system according to an embodiment is shown in

FIG. 2A. The system includes a DC voltage/signal generator **23** that is driven by a controller unit **21** that interfaces with a computer device **24** by means of a two-way communication link **29**. The controller can perform channel selection and routing functions for applying DC voltages to appropriate electrodes that have been selected by a user or by the computer **24**, and apply the voltages via a multiplicity of leads (shown collectively as **26**) to a catheter device **22**. The catheter device can be any of the catheters shown and described herein or in the '394 PCT Application. Some leads from the controller **21** could also carry pacing signals to drive pacing of the heart through a separate pacing device (not shown). The catheter device can also send back information such as ECG (electrocardiograph) recordings or data from other sensors back to the controller **21** as indicated by the data stream **25**, possibly on separate leads. While the DC voltage generator **23** sends a DC voltage to the controller **21** through leads **27**, the voltage generator is driven by control and timing inputs **28** from the controller unit **21**.

(46) In some embodiments, the electrode controller can include one or more modules and can automatically perform channel selection (e.g., identification of a subset of electrodes to which the voltage pulses will be applied), identification of the desired anode/cathode pairs, or the like. For example, FIG. 2B shows an electroporation system according to an embodiment that includes an electrode controller **900** and a signal generator **925**. The electrode controller **900** is coupled to a computer **920** or other input/output device, and is configured to be operable coupled to a medical device **930**. The medical device **930** can be one or more multi-electrode catheters, of the types shown and described herein. Further the medical device **930** can be coupled to, disposed about and/or in contact with a target tissue, such as the heart H. In this manner, as described herein, the electroporation system, including the electrode controller **900** and the signal generator **925**, can deliver voltage pulses to the target tissue for therapeutic purposes.

(47) The controller **900** can include a memory **911**, a processor **910**, and an input/output module (or interface) **901**. The controller **900** can also include a pacing module **902**, a feedback module **905**, a pulse delivery module **908**, and a selection module **912**. The electrode controller **900** is coupled to a computer **920** or other input/output device via the input/output module (or interface) **901**.

(48) The processor **910** can be any processor configured to, for example, write data into and read data from the memory **911**, and execute the instructions and/or methods stored within the memory **911**. Furthermore, the processor **910** can be configured to control operation of the other modules within the controller (e.g., the pacing module **902**, the feedback module **905**, the pulse delivery module **908**, and the selection module **912**). Specifically, the processor can receive a signal including user input, impedance, heart function or the like information and determine a set of electrodes to which voltage pulses should be applied, the desired timing and sequence of the voltage pulses and the like. In other embodiments, the processor **910** can be, for example, an application-specific integrated circuit (ASIC) or a combination of ASICs, which are designed to perform one or more specific functions. In yet other embodiments, the microprocessor can be an analog or digital circuit, or a combination of multiple circuits.

(49) The memory device **910** can be any suitable device such as, for example, a read only memory (ROM) component, a random access memory (RAM) component, electronically programmable read only memory (EPROM), erasable electronically programmable read only memory (EEPROM), registers, cache memory, and/or flash memory. Any of the modules (the pacing module **902**, the feedback module **905**, the pulse delivery module **908**, and the selection module **912**) can be implemented by the processor **910** and/or stored within the memory **910**.

(50) As shown, the electrode controller **900** operably coupled to the signal generator **925**. The signal generator includes circuitry, components and/or code to produce a series of DC voltage pulses for delivery to electrodes included within the medical device **930**. For example, in some embodiments, the signal generator **925** can be configured to produce a biphasic waveform having a pre-polarizing pulse followed by a polarizing pulse. The signal generator **925** can be any suitable signal generator of the types shown and described herein.

(51) The pulse delivery module **908** of the electrode controller **900** includes circuitry, components and/or code to deliver an output signal associated with the pulsed voltage waveform produced by the signal generator **925**. This signal (shown as signal **909**) can be any signal of the types shown and described herein, and can be of a type and/or have characteristics to be therapeutically effective. In some embodiments, the pulse delivery module **908** receives input from the selection module **912**, and can therefore send the signal **909** to the appropriate subset of electrodes, as described herein.

(52) The selection module **912** includes circuitry, components and/or code to select a subset of electrodes from the electrodes included within the medical device **930**, as described herein. In some embodiments, the selection module **912** is configured identify at least one electrode from the subset of electrodes as an anode and at least one electrode from the subset of electrodes as a cathode. In some embodiments, the selection module **912** is configured to select a subset of electrodes from more than one medical device **930**, as described herein.

(53) In some embodiments, the selection module **912** can select the subset of electrodes based on input received from the user via the input/output module **901**. For example, in some embodiments, the user can use visualization techniques or other methods to identify the desired electrodes, and can manually enter those electrodes, as described herein.

(54) In other embodiments, the selection module **912** is configured to select the subset of electrodes based on a predetermined schedule of the set of electrodes. For example, in some embodiments, the electrode controller **900** can be configured accommodate different medical devices **930** having different numbers and/or types of electrodes. In such embodiments, the selection module **912** can retrieve a predetermined schedule of electrodes to which a series of voltage pulses can be applied, based on the specific type of medical device **930**.

(55) In yet other embodiments, the selection module **912** is configured to select the subset of electrodes automatically based on at least one of an impedance associated with the subset of electrodes, a distance between the first electrode and the second electrode, and a characteristic associated with a target tissue. For example, in some embodiments, the electrode controller **900** includes the feedback module **905** that can receive feedback from the medical device **930** (as identified by the signal **906**). The feedback module **905** includes circuitry, components and/or code to determine an impedance between various electrodes (as described herein). Thus, in such embodiments, the selection module **912** can select the subset of electrodes automatically based the impedance.

(56) In some embodiments, the electrode controller **900** optionally includes the pacing module **902**. The pacing module **902** includes circuitry, components and/or code to produce a pacing signal (identified as signal **903**) that can be delivered to the target tissue (or heart) via a pacing lead. As described herein, the pacing module **902** can facilitate any suitable “mode” of operation desired, such as a standard pacing, an overdrive pacing option (for pacing at a faster-than-normal heart rate), an external trigger option (for pacing from an externally generated trigger), and a diagnostic pacing option.

(57) As shown in FIG. 3, given atrial or ventricular pacing inputs to the heart (e.g., from the pacing module **902** of the electrode controller **900**), the resulting ECG waveform **32** has appropriate respective refractory time intervals **33** and **34** respectively, during which there are suitable time windows for application of irreversible electroporation as indicated by **35** and **36**. The application of cardiac pacing results in a periodic, well-controlled sequence of electroporation time windows. Typically, this time window is of the order of hundreds of microseconds to about a millisecond or more. During this window, multiple DC voltage pulses can be applied to ensure that sufficient tissue ablation has occurred. The user can repeat the delivery of irreversible electroporation over several successive cardiac cycles for further confidence. Thus, in some embodiments, a feedback module (e.g., feedback module **905**) can receive the electrocardiogramal, and a pulse delivery module (e.g., pulse delivery module **908**) can deliver the output signal to the subset of electrodes

during a time window associated with at least one a pacing signal or the electrocardiograph signal.

(58) In some embodiments, the ablation controller and signal generator can be mounted on a rolling trolley, and the user can control the device using a touchscreen interface that is in the sterile field. Referring to FIG. 2B, in such embodiments, the computer **920** can be a touchscreen device. The touchscreen can be for example an LCD touchscreen in a plastic housing mountable to a standard medical rail or post and can be used to select the electrodes for ablation and to ready the device to fire. The interface can for example be covered with a clear sterile plastic drape. The operator can select the number of electrodes involved in an automated sequence. The touch screen graphically shows the catheters that are attached to the controller. In one embodiment the operator can select electrodes from the touchscreen with appropriate graphical buttons. The operator can also select the pacing stimulus protocol (either internally generated or externally triggered) from the interface. Once pacing is enabled, and the ablation sequence is selected, the operator can initiate or verify pacing. Once the operator verifies that the heart is being paced, the ablation sequence can be initiated by holding down a hand-held trigger button that is in the sterile field. The hand-held trigger button can be illuminated red to indicate that the device is “armed” and ready to ablate. The trigger button can be compatible for use in a sterile field and when attached to the controller can be illuminated a different color, for example white. When the device is firing, the trigger button flashes in sequence with the pulse delivery in a specific color such as red. The waveform of each delivered pulse is displayed on the touchscreen interface. A graphic representation of the pre and post impedance between electrodes involved in the sequence can also be shown on the interface, and this data can be exported for file storage.

(59) In some embodiments, an impedance map can be generated based on voltage and current recordings across anode-cathode pairs or sets of electrodes, and an appropriate set of electrodes that are best suited for ablation delivery in a given region can be selected based on the impedance map or measurements, either manually by a user or automatically by the system. Such an impedance map can be produced, for example, by the feedback module **905**, or any other suitable portion of the electrode controller **900**. For example, if the impedance across an anode/cathode combination of electrodes is a relatively low value (for example, less than 25 Ohms), at a given voltage the said combination would result in relatively large currents in the tissue and power dissipation in tissue. In such circumstances, this electrode combination would then be ruled out (e.g., via the selection module **912**) for ablation due to safety considerations, and alternate electrode combinations would be sought by the user. In a preferred embodiment, a pre-determined range of impedance values, for example 30 Ohms to 300 Ohms, could be used as an allowed impedance range within which it is deemed safe to ablate. Thus, in some embodiments, an electrode controller can automatically determine a subset of electrodes to which voltage pulses should be applied.

(60) The waveforms for the various electrodes can be displayed and recorded on the case monitor and simultaneously outputted to a standard connection for any electrophysiology (EP) data acquisition system. With the high voltages involved with the device, the outputs to the EP data acquisition system needs to be protected from voltage and/or current surges. The waveforms acquired internally can be used to autonomously calculate impedances between each electrode pair. The waveform amplitude, period, duty cycle, and delay can all be modified, for example via a suitable Ethernet connection. Pacing for the heart is controlled by the device and outputted to the pacing leads and a protected pacing circuit output for monitoring by a lab.

(61) While a touchscreen interface is one preferred embodiment, other user interfaces can be used to control the system such as a graphical display on a laptop or monitor display controlled by a standard computer mouse or joystick. FIG. 4 shows a schematic rendering of a portion of the user interface of the electroporation system. The four windows A, B, C and D shown in the FIG. represent four different choices of electrode subsets for anode and cathode selection. The PV isolation ablation catheter is represented by a string of numbered electrodes as indicated by **41**, wrapped around the area **42** of the pulmonary veins represented by the enclosed region in this

schematic diagram. Referring to B in the FIG., the dashed-line vectors **46** represent approximate current vectors, with their tips at the cathodes and tails or bases at the anodes; in this FIG., the three electrodes marked **43** are cathodes, and the single electrode marked **44** is the anode. The windows A, C and D in this FIG. show other choices of cathode and anode electrode subsets, in each case with accompanying approximate current density vectors shown by dashed arrows. It is clear from the FIG. that the user can select various subsets of electrodes as cathode or anode, depending on the region to be ablated along the length of the PV isolation catheter. In one embodiment, the user can make one selection of cathode and anode subsets, and the system can take this selection as input to generate an ablation sequence that moves around the ring or contour defined by the shape of the PV isolation catheter, for example moving clockwise at each step with a one-electrode displacement. In this manner, the pair of cathode and anode electrode subsets can be sequentially updated for ablation purposes, so that if there are N electrodes, after N updates the entire contour has been updated such that the tips of the current arrows shown as **46** have swept once around the contour completely.

(62) In some cases, the portion of the PV isolation catheter with electrodes may be longer than needed to wrap around a given patient's pulmonary veins; in this event, a smaller number of electrodes suffices to wrap around the contour of the pulmonary veins. These define the number of “active” electrodes to be used in the ablation process. In the specific example shown in FIG. 5 for a 30-electrode PV isolation catheter, the set of electrodes wrapped around the pulmonary veins, represented by **51**, are the electrodes that would be used in the ablation process, and are called out by the label **50** as the set of active electrodes, while the remaining five electrodes represented by **52** are not used in the ablation process as they are not located in suitable positions.

(63) In one embodiment of the electroporation system disclosed herein, multiple catheters could be used for ablation, with anode and cathode subsets respectively chosen to lie on different catheters. In one exemplary use of such multi-device electroporation, two catheters are connected to the controller unit of the electroporation system. In some instances of ablation procedures for the treatment of Atrial Fibrillation (AF), in addition to isolating the pulmonary veins, it is also useful to generate an ablation line that separates or isolates regions around the mitral valve. Such a line is often termed a “Mitral Isthmus Line,” and is a line running from an Inferior aspect of the PV isolation contour to the outer edge of the mitral valve in the left atrium. In some embodiments, a method of use of the systems described herein includes inserting a coronary sinus catheter with multiple electrodes into the coronary sinus. As shown in FIG. 6, electrodes marked as **66** on the coronary sinus catheter **63**, and electrodes marked as **65** on the PV isolation catheter **62**, are further selected as active subsets of electrodes for ablation, as indicated in the indication window **61** for the anode and cathode subsets. Thus in effect the electrodes of the coronary sinus catheter **63** are utilized to define the mitral isthmus line, and irreversible electroporation ablation can be effectively performed on the left atrial wall to generate a mitral isthmus ablation line.

(64) In some embodiments, the system (any of the generators and controllers described herein) can deliver rectangular-wave pulses with a peak maximum voltage of about 5 kV into a load with an impedance in the range of 30 Ohm to 3000 Ohm for a maximum duration of 200 μ s, with a 100 μ s maximum duration being still more preferred. Pulses can be delivered in a multiplexed and synchronized manner to a multi-electrode catheter inside the body with a duty cycle of up to 50% (for short bursts). The pulses can generally be delivered in bursts, such as for example a sequence of between 2 and 10 pulses interrupted by pauses of between 1 ms and 1000 ms. The multiplexer controller is capable of running an automated sequence to deliver the impulses/impulse trains (from the DC voltage signal/impulse generator) to the tissue target within the body. The controller system is capable of switching between subsets/nodes of electrodes located on the single use catheter or catheters (around or within the heart). Further, the controller can measure voltage and current and tabulate impedances in each electrode configuration (for display, planning, and internal diagnostic analysis). It can also generate two channels of cardiac pacing stimulus output, and is capable of

synchronizing impulse delivery with the internally generated cardiac pacing and/or an external trigger signal. In one embodiment, it can provide sensing output/connection for access to bio potentials emanating from each electrode connected to the system (with connectivity characteristics being compatible with standard electrophysiological laboratory data acquisition equipment).

(65) In some embodiments, the controller (e.g., the controller **900**) can automatically “recognize” the single-use disposable catheter when it is connected to the controller output (prompting internal diagnostics and user interface configuration options). The controller can have at least two unique output connector ports to accommodate up to at least two catheters at once (for example, one 30-electrode socket and one 10-electrode socket; a 2-pole catheter would connect to the 10-pole socket). The controller device can function as long as at least one recognized catheter is attached to it. In a preferred embodiment, the controller can have several sequence configurations that provide the operator with at least some variety of programming options. In one configuration, the controller can switch electrode configurations of a bipolar set of electrodes (cathode and anode) sequentially in a clockwise manner (for example, starting at step n , in the next step of the algorithm, cathode $n+1$ and anode $n+1$ are automatically selected, timed to the synchronizing trigger). With the 30-pole catheter the electrodes are arranged in a quasi-circumference around the target. Thus in the first sequence, pulse delivery occurs so that the vector of current density changes as the automated sequencing of the controller switches “on” and “off” between different electrodes surrounding the tissue target sequence. The current density vectors generally cross the target tissue but in some configurations the current density could be approximately tangential to the target. In a second sequence configuration, the impulses are delivered to user-selected electrode subsets of catheters that are connected to the device (the vector of current density does not change with each synchronized delivery). A third sequence configuration example is a default bipolar pulse sequence for the simplest 2-pole catheter. The user can also configure the controller to deliver up to 2 channels of pacing stimulus to electrodes connected to the device output. The user can control the application of DC voltage with a single handheld switch. A sterile catheter or catheters can be connected to the voltage output of the generator via a connector cable that can be delivered to the sterile field. In one embodiment, the user activates the device with a touch screen interface (that can be protected with a single-use sterile transparent disposable cover commonly available in the catheter lab setting). The generator can remain in a standby mode until the user is ready to apply pulses at which point the user/assistant can put the generator into a ready mode via the touchscreen interface. Subsequently the user can select the sequence, the active electrodes, and the cardiac pacing parameters.

(66) Once the catheter has been advanced to or around the cardiac target, the user can initiate electrically pacing the heart (using a pacing stimulus generated by the ablation controller or an external source synchronized to the ablation system). The operator verifies that the heart is being paced and uses the hand-held trigger button to apply the synchronized bursts of high voltage pulses. The system can continue delivering the burst pulse train with each cardiac cycle as long as the operator is holding down a suitable “fire” button or switch. During the application of the pulses, the generator output is synchronized with the heart rhythm so that short bursts are delivered at a pre-specified interval from the paced stimulus. When the train of pulses is complete, the pacing continues until the operator discontinues pacing.

(67) FIG. 7 shows a portion of a graphical user interface according to an embodiment for focal ablation with a bipolar or 2-electrode catheter. In some embodiments, the graphical user interface can be a touchscreen interface. A graphical icon **71** for a bipolar catheter is shown along with a depth selection button **72** for selecting a desired ablation depth. In one embodiment of the system, based on the latter depth setting, the system can select an appropriate voltage value to apply in order to ensure that an electric field sufficient to cause irreversible electroporation is applied up to the desired depth.

(68) In some embodiments, a pacing selection interface on a portion of the user interface of the

electroporation system is shown in FIG. 8. The pacing selection interface has various options such as an overdrive pacing option **81** for pacing at a faster-than-normal heart rate, an external trigger option **82** for pacing from an externally generated trigger, and a diagnostic pacing option **83**. By clicking the “Select” button for a given option, that option is selected.

(69) As an example of a pacing option selected, FIG. 9 shows a portion the user interface of the electroporation system with pacing from an external trigger selected for the pacing option. The selected option **86** is displayed along with the pacing characteristics and waveform **87** for the user to visualize.

(70) FIG. 10 shows a portion the user interface of the electroporation system for selection of anode and cathode electrodes, with two catheters connected to the system. One of the catheters is a PV isolation catheter **91** while the other is a multi-electrode catheter **92**. The buttons **93** and **94** can implement the selection of marked electrode subsets on the catheters as respectively anode or cathode. Once the selection is made, the appropriate electrodes are colored differently to indicate anode or cathode electrode as shown marked respectively as **96** and **97** in FIG. 11.

(71) The controller and generator can output waveforms that can be selected to generate a sequence of voltage pulses in either monophasic or biphasic forms and with either constant or progressively changing amplitudes. FIG. 12 shows a rectangular wave pulse train where the pulses **101** have a uniform height or maximum voltage. FIG. 13 shows an example of a balanced biphasic rectangular pulse train, where each positive voltage pulse such as **103** is immediately followed by a negative voltage pulse such as **104** of equal amplitude and opposite sign. While in this example the biphasic pulses are balanced with equal amplitudes of the positive and negative voltages, in other embodiments an unbalanced biphasic waveform could also be used as may be convenient for a given application.

(72) Yet another example of a waveform or pulse shape that can be generated by the system is illustrated in FIG. 14, which shows a progressive balanced rectangular pulse train, where each distinct biphasic pulse has equal-amplitude positive and negative voltages, but each pulse such as **107** is larger in amplitude than its immediate predecessor **106**. Other variations such as a progressive unbalanced rectangular pulse train, or indeed a wide variety of other variations of pulse amplitude with respect to time can be conceived and implemented by those skilled in the art based on the teachings herein.

(73) The time duration of each irreversible electroporation rectangular voltage pulse could lie in the range from 1 nanosecond to 10 milliseconds, with the range 10 microseconds to 1 millisecond being more preferable and the range 50 microseconds to 300 microseconds being still more preferable. The time interval between successive pulses of a pulse train could be in the range of 10 microseconds to 1 millisecond, with the range 50 microseconds to 300 microseconds being more preferable. The number of pulses applied in a single pulse train (with delays between individual pulses lying in the ranges just mentioned) can range from 1 to 100, with the range 1 to 10 being more preferable. As described in the foregoing, a pulse train can be driven by a user-controlled switch or button, in one embodiment preferably mounted on a hand-held joystick-like device. In one mode of operation a pulse train can be generated for every push of such a control button, while in an alternate mode of operation pulse trains can be generated repeatedly during the refractory periods of a set of successive cardiac cycles, for as long as the user-controlled switch or button is engaged by the user.

(74) In one embodiment of a biphasic waveform, a brief pre-polarization pulse can be applied just prior to the application of a polarizing rectangular pulse. The rapid change in electric field in tissue, in addition to the electric field magnitude, driven by this type of pulse application incorporating a pre-polarizing pulse can promote a more rapid and effective tissue ablation in some applications. A schematic diagram of a voltage/signal generator for the purpose of generating such a waveform employing intrinsic amplification of voltage spikes arising from switching is given in FIG. 15. A DC voltage is generated from the discharge of a suitable capacitor bank **111** (powered by a suitable

charging circuit that is not shown and that would be familiar to those skilled in the art) with a suitable high-voltage diode or rectifier **112** serving to ensure voltage polarity. While in one embodiment of the signal generator the voltage discharge passes initially through a snubber or transient-suppression circuit **113** that strongly suppresses transients, in an alternate embodiment an alternate signal path is available via a switch unit **116** that detects and lets through a brief initial voltage spike after a possible voltage inversion. The signal then passes through an Insulated Gate Bipolar Transistor (IGBT) high-power switch **115** and is accessible at terminals **117** for connection to catheter electrodes. When no pre-polarizing pulse is required, the switch unit **116** would not be present, and the signal passes (after transient-suppression in the snubber circuit **113**) initially through a Metal Oxide Semiconductor Field Effect Transistor (MOSFET) switch **114** before passing through the IGBT switch **115**. The IGBT interfaces better with a wider range of load impedances (from tissue), while the faster switching speed of the MOSFET can drive suitably fast turn-on/turn-off of the IGBT switch, so that using them in the sequence shown in FIG. **15** can be advantageous.

(75) In order to generate a sequence of rectangular pulses, the time constant associated with the capacitor bank discharge is chosen to be significantly longer than an individual pulse duration. If the charging circuit of the capacitor bank is much more rapid, a sequence of very highly rectangular pulses can be generated from repeated capacitor bank discharges. As shown in FIG. **16**, when a capacitor bank is suddenly discharged by closing a switch, the voltage can briefly be amplified and spike as indicated by **122** before settling to a normal discharge pattern **121**. The inductance of leads connected to the capacitor forms a tank circuit with the capacitor, and in many cases this tank circuit is rapidly driven to resonance by the discharge, leading to the voltage amplification seen in the spiking behavior **122**. Indeed, by suitably controlling lead inductance and resistance, the extent of spike generated (spike amplitude and duration) can be further controlled. Such additional inductance/resistance or spike control circuitry can be included in the switch unit **116** shown in FIG. **15**. Thus the switch **116** plays a gatekeeping role akin to the notion of “Maxwell's demon” in statistical physics, allowing the passage of some types of intrinsically amplified voltage spikes, and accordingly the signal generator of FIG. **15** can also be termed a Maxwell amplifier.

(76) A schematic representation of a biphasic pulse with pre-polarization is shown in FIG. **17**, where each rectangular polarizing pulse such as **125** is preceded by a pre-polarizing negative voltage spike **126** derived from transient spiking from a generator of the type shown in FIG. **15**. The value of the negative spike voltage peak V_L could be determined by the spike control circuitry of the switch unit **116** as discussed above. The biphasic pulses involve a time duration T_i for the pre-polarizing spike, a duration T_2 for the rectangular polarizing pulse, with a time delay T_a indicated by **127** in FIG. **17** between successive biphasic pulses. Typical values of T_2 could be of the order of 100 microseconds, while T_i could lie in the range 5 microseconds to 50 microseconds and the delay time T_a could be in the range 100 microseconds to 300 microseconds. All of these parameters can be determined by the design of the signal generator, and in various embodiments could also be determined by user control as may be convenient for a given clinical application. The specific examples and descriptions herein are exemplary in nature and variations can be developed by those skilled in the art based on the material taught herein without departing from the scope of the embodiments described herein.

(77) A catheter device for focal ablation with the electroporation system according to an embodiment is shown schematically in FIG. **18A**. The focal ablation catheter **131** has two electrodes disposed in the distal section of the catheter, with a relatively proximally placed cathode electrode **133** of length L_1 exposed on the catheter shaft and a relatively distally placed anode electrode **134** of length L_2 mounted on the inner section of the shaft. The catheter shaft is made of a material with high dielectric strength such as for example a polymer comprising Teflon. Thus the distal electrode **134** is covered by the polymeric shaft and is not exposed to the blood pool on the

shaft exterior. Both electrodes are metallic, and in one embodiment the anode could be poly-metallic in construction, for example comprising regions of Titanium and regions of Platinum. (78) As shown in FIG. **18B**, the anode **141** is recessed from the distal tip, and could be placed with its distal portion between 0.5 mm and 10 mm away from the distal end of the catheter. While the recessed and interior placement of the anode electrode is counterintuitive, it can be an effective means of enhancing electroporation efficacy and safety selectively for focal ablation. As indicated in the FIG., the schematically depicted (negative) current density flowing from the cathode to the anode indicated by the stream of arrows **142** bends around the distal edge **143** of the catheter shaft resulting in a region of high curvature of the current density, and correspondingly curved electric field lines as well. If the anode electrode is recessed, the highest curvature region at the edge of the electrode where the strongest electric field occurs is also displaced proximally, so that the electric field around the distal tip of the catheter itself is not so large as to cause any increased local heating, while remaining large enough to cause irreversible electroporation. FIG. **18C** schematically shows electric field lines **152** as they curve to enter the distal catheter end **151**, resulting in a zone or region **153** distal to the catheter with a relatively uniform electric field. The electric field in this region would not be as uniform with a standard, externally placed electrode on the catheter, and further would generally have regions of very high intensity. The recessed interior placement of the distal electrode therefore results in superior and safe electroporation ablation delivery for focal ablation in a selective zone distal to the catheter tip. While in one embodiment the anode electrode can be a hollow ring, in other embodiments it could have a different form such as a cylinder with multiple longitudinal holes or channels.

(79) In some embodiments, the interior lumen of the catheter can carry a fluid delivered out through the distal end of the catheter. The fluid can be an ionic fluid such as isotonic or hypertonic saline and can enhance electrical conductivity in the distal region of the catheter and beyond the distal end ensuring a proper and more uniform distribution of local electric field in a distal region around the catheter. As shown in FIG. **18D**, the inner lumen **163** of the catheter carries a flowing fluid indicated by the arrows, while the anode **161** is in the shape of a cylinder that has longitudinal channels through which fluid can flow. The anode electrode is shown in cross section **162** where the multiple through-channels are visible. The saline or ionic fluid delivered through the channels can act as an ion bridge through which good electrical current conduction is possible in the vicinity of the catheter's distal end, resulting in a more uniform distribution of local electric field and electroporation energy delivery. Further, the flow of saline fluid itself can dislodge tissue debris or any bubbles generated by the electroporation-driven breakdown of tissue in the region around the distal catheter tip. In one embodiment, ultrasonic waves schematically shown as **164** in FIG. **18D** can be applied to the fluid to further ensure that any bubbles that may be lodged on the catheter surface in its distal region are dislodged, again for purposes of maintaining relative uniformity of the local electric field. The ultrasonic waves or ultrasound through the fluid can be generated by suitable piezoelectric transducers mounted more proximally along the shaft of the catheter, or even within the catheter handle. In one embodiment, ultrasound generation is coordinated with electroporation DC voltage application so that the kinetic energy of focused ultrasound is available "on demand."

(80) The above statements of the advantages of the recessed inner electrode for the focal ablation catheter have been verified by the inventors in physically realistic simulations. FIG. **19A** shows a simulation geometry of a catheter **601** in a blood pool region **602** and abutting a myocardium region **603** adjacent to a pericardial region **605**, all regions being assumed cylindrical for the simulation; the section or plane **611** was chosen to plot electric field intensity contours. Further the myocardium region has a region of scar tissue **604** distal to the catheter. FIG. **19B** shows the catheter's distal region with distal tip **612**, distal (anode) recessed electrode **615** mounted internal to the shaft, and a proximal (cathode) electrode **616**. With a DC voltage of 1000 Volts applied between the electrodes and using realistic physical property values, the resulting electric field

intensity contours are illustrated in FIG. 19C, which shows the geometry and the contours in the section **611** of FIG. 19A. The cathode **616** and recessed anode **615** are indicated in FIG. 19C, as are the blood pool region **602**, the myocardium region **603** and the pericardial region **605**. The electric field intensity contours corresponding to 1000 Volts/cm, 500 Volts/cm and 200 Volts/cm respectively are shown as contours **617**, **618** and **619** respectively.

(81) FIG. 19D repeats the electric field simulation for a catheter with both electrodes mounted externally, showing the distal anode **622** and proximal cathode **623** in the same tissue geometry. As before, with a DC voltage of 1000 Volts applied between the electrodes and using the same realistic physical property values, the resulting electric field intensity contours corresponding to 1000 Volts/cm, 500 Volts/cm and 200 Volts/cm respectively are shown as contours **626**, **627** and **628** respectively.

(82) Comparing the intensity contours of FIG. 19C and FIG. 19D, it is clear that the former FIG. shows a relatively more uniform intensity distribution in the myocardium region where excessively large electric field values (500 Volts/cm or more) are present to a much lesser extent, mitigating the possibility of locally large electric current densities and corresponding temperature increases.

(83) The focal ablation catheter described above in various embodiments could be used in cardiac applications such as ablation delivery to treat Ventricular Tachycardia (VT), where targeted ablation delivery could be of great benefit. In one embodiment, the length L2 of the distal anode electrode could be significantly longer than the length L1 of the proximal cathode electrode. The ratio L2/L1 could have a value of at least 1.3, more preferably lie in the range 1.3 to 10, and still more preferably in the range 2 to 5. The increased surface area of the exposed inner surface of the anode electrode can serve to reduce the current density near it, thereby enhancing safety and enhancing the efficacy of ion bridging current transfer with the saline fluid infusion. In addition or as an alternate method to reduce high current density due to exposed metal regions with high curvature, the edges of the electrode can be beveled or rounded to ensure that there are no sharp corners or regions with high curvature.

(84) Thus, the methods described herein allow for a variety of approaches in the context of cardiac ablation. Considering for example the treatment of Ventricular Tachycardia (VT) as a clinical application, electroporation ablation of cardiac tissue could be performed across a pair of nearby electrodes respectively on one or more epicardially placed catheters in one embodiment. In an alternate embodiment, a single focal ablation endocardial catheter such as described above can be used to ablate on the endocardial side of a cardiac chamber. In still another alternate embodiment, a bipolar pair of electrodes on different catheters, one placed endocardially and the other epicardially could be used to drive irreversible electroporation.

(85) A balloon ablation device for use with the electroporation system according to an embodiment is schematically illustrated in FIG. 20. While the device is shown to be situated in a blood vessel **301**, it can also be used in other anatomical areas such as a cardiac ventricle. The device shaft **302** has an inflatable balloon **306** disposed in its distal portion. On either side of the balloon and mounted on the shaft are a proximal cathode electrode **303** and a distal anode electrode **304**. The balloon surface **305** is coated with a thin layer of a good insulator such as a biocompatible metal oxide (for example, aluminum oxide). When air is pumped in through the catheter shaft lumen and through appropriate openings (not shown) from the shaft into the balloon, the balloon can inflate, as the inflated shape in FIG. 20 depicts. In ventricular applications such as cardiac ablation for VT, the balloon can serve to displace collateral structures away from the distal end of the catheter. Since the balloon surface is an insulator, when a DC voltage is applied between the electrodes, the current flow between electrodes and through the tissue is deflected around the balloon. The electric field is also correspondingly deflected around the surface of the balloon, and curves outward and can extend into the wall of the blood vessel, as shown by the schematic electric field lines **308** in the FIG. In one embodiment, the distal electrode **304** can be mounted on the outside of the shaft with external surface exposed, for intravascular ablation applications such as for example peripheral

vascular applications for treatment of atherosclerosis where it is desired to ablate the vessel wall region or clear deposits. In an alternate embodiment, the distal electrode **304** can be mounted on the inner side of the catheter shaft, so that the metal electrode is internally exposed. Furthermore, the distal electrode could be recessed from the distal tip, as schematically shown in FIG. **20**. Such a device could be used in focal ablation applications.

(86) An alternate preferred embodiment of a balloon ablation device for use with the electroporation system according to an embodiment is shown schematically in FIG. **21A**. As before, while the device is shown to be situated in a blood vessel **172**, it can also be used in other anatomical areas such as a cardiac ventricle. The device shaft **171** has an inflatable balloon **177** disposed in its distal portion. On either side of the balloon and mounted on the shaft are a proximal cathode electrode **173** and a distal anode electrode **174**. The balloon surface **176** is coated with a thin layer of a good insulator such as a biocompatible metal oxide (for example, aluminum oxide). When fluid is pumped in through the catheter shaft lumen, the balloon can inflate, as the inflated shape in FIG. **21A** depicts. The region of the shaft within the balloon can have at least one ultrasonic transducer **175** mounted thereon. Further, while the interior of the balloon is inflated with fluid, the wall **178** of the balloon has a varying cross section and is filled with a gas such as air. Thus there is a liquid-gas interface at the wall of the balloon. This interface can act as a mirror or reflecting surface for ultrasound. As illustrated in the FIG., ultrasound rays **179** emitted by the transducer **175** are reflected at the wall of the balloon and are subsequently focused on a ring-like region, shown with its annular regions **181** in a plane **180** and viewed edge-on in the illustration. The annular region external to the balloon where the ultrasound is focused is shown in FIG. **21B** as the dark annulus **203** surrounding the vessel wall **201**, shown in cross section. When high-intensity ultrasound generated by the transducer in the catheter is focused in this manner, its energy is deposited in the annulus as heat and raises temperature locally. The irreversible electroporation threshold of tissue is temperature dependent and is lowered with increasing temperature. Thus in the annular region where temperature is increased due to the focused ultrasound, the electroporation threshold is decreased, and a relatively weaker electric field in this region can still generate irreversible electroporation, permitting selective tissue ablation.

(87) As before, in ventricular applications such as cardiac ablation for VT, the balloon can serve to displace collateral structures away from the distal end of the catheter. Since the balloon surface is an insulator, when a DC voltage is applied between the electrodes, the current flow between electrodes and through the tissue is deflected around the balloon. The electric field is also correspondingly deflected around the surface of the balloon, and curves outward and can extend into the wall of the blood vessel, as shown by the schematic electric field lines **191** in the FIG. With the lowered irreversible electroporation threshold in the annular region of focused ultrasound in the vessel wall, the electric field in the annular region is sufficient to selectively drive irreversible electroporation. In one embodiment, the distal electrode **174** can be mounted on the outside of the shaft with external surface exposed, for intravascular ablation applications such as for example peripheral vascular applications for treatment of atherosclerosis where it is desired to ablate the vessel wall region or clear deposits. In an alternate embodiment, the distal electrode **174** can be mounted on the inner side of the catheter shaft, so that the metal electrode is internally exposed. Furthermore, the distal electrode could be recessed from the distal tip, as schematically shown in FIG. **21A**. Such a device could be used in focal ablation applications.

(88) While the specific embodiment of the balloon ablation device described above utilizes high-intensity focused ultrasound to selectively generate temperature increases in a given region in order to decrease the electroporation threshold, it must be noted that alternate energy delivery mechanisms such as microwaves could be utilized for the purpose of increasing tissue temperature by energy deposition in order to decrease the irreversible electroporation threshold electric field. The balloon ablation devices described in the foregoing could also be used in pulmonary outflow tract applications to treat pulmonary hypertension, or in esophageal or gastrointestinal

applications where tissue ablation is an appropriate therapy.

(89) An ablation device for irreversible electroporation in the form of an expanding basket catheter is schematically illustrated in FIG. 22. Mounted on the shaft **501** of the device is a sliding member **505** which is attached to a system of struts **507**. While the system of struts is initially substantially aligned with the length of the shaft in a folded configuration (not shown), it can unfold or open out like an umbrella by movement of the sliding member **505** along the shaft, resulting in the basket-like unfolded structure shown in FIG. 22. For purposes of clarity, the system of struts is only partially shown in the figure. The basket construction or set of expanding struts can comprise a superelastic alloy such as for instance an alloy of Nickel and Titanium often called Nitinol®; such a construction can be of use in a large vessel where a significant amount of expansion could be called for. On either side of the basket-like structure and disposed along the shaft are two electrodes **502** and **503**. Bead-like electrodes **508** are present at corners of the strut assembly. The device be guided to a desired location inside a large vascular or other anatomical vessel (for example, the pulmonary outflow tract), positioned suitably and then unfolded. Subsequently, irreversible electroporation ablation can be applied with the electrodes on the basket device.

(90) In one embodiment the basket catheter with struts folded can have a lumen for passage of a suitable guidewire which could be used as a delivery system for appropriate placement of the basket catheter. Various choices of electrode configurations for ablation are possible for this device. In one preferred embodiment, either of electrodes **502** or **503** is selected as cathode, while the beads **508** are selected as anodes. With such a choice, a region of vessel wall that is located just proximal to or just distal to the beads can be selectively ablated, depending on whether electrode **502** or electrode **503** respectively is activated as cathode. The rounded shape of the beads and their location on the outer edge of the basket (and thus close to the vessel wall) results in good ablation characteristics at the vessel wall. The voltage applied can be suitably selected so that an appropriate electric field is generated in the ablative region near the bead electrodes. In alternate preferred embodiments, the basket catheter can further incorporate energy sources such as focused ultrasound or microwaves in order to selectively raise temperatures in localized regions and thereby lower the irreversible electroporation threshold.

(91) In some embodiments, a method includes identifying, via a selection module of an electrode controller, a set of anode/cathode pairs from a set of electrodes of a multi-electrode catheter. The method can be performed using any suitable controller, such as for example, the controller **900** described above. The multi-electrode catheter is configured to be disposed about a portion of a heart. At least one of the anode/cathode pair including at least one anode electrode and at least one cathode electrode. In other embodiments, however, the anode/cathode pair can include multiple anode electrodes or cathode electrodes. In some embodiments, the identifying can be based on input received from an input/output module of the electrode controller (e.g., manual input). In other embodiments, the identifying can be based on a predetermined schedule of electrodes. In yet other embodiments, the identifying can be performed automatically based on an impedance map as described herein.

(92) The method further includes conveying a pacing signal to a pacing lead configured to be operatively coupled to the heart, and receiving, at a feedback module of the electrode controller, an electrocardiogram associated with a function of the heart.

(93) The method further includes delivering, via a pulse delivery module of the electrode controller, a pulsed voltage waveform to the plurality of anode/cathode pairs according to a sequential pattern.

(94) Some embodiments described herein relate to a computer storage product with a non-transitory computer-readable medium (also can be referred to as a non-transitory processor-readable medium) having instructions or computer code thereon for performing various computer-implemented operations. The computer-readable medium (or processor-readable medium) is non-transitory in the sense that it does not include transitory propagating signals per se (e.g., a propagating electromagnetic wave carrying information on a transmission medium such as space or

a cable). The media and computer code (also can be referred to as code) may be those designed and constructed for the specific purpose or purposes. Examples of non-transitory computer-readable media include, but are not limited to: magnetic storage media such as hard disks, floppy disks, and magnetic tape; optical storage media such as Compact Disc/Digital Video Discs (CD/DVDs), Compact Disc-Read Only Memories (CD-ROMs), and holographic devices; magneto-optical storage media such as optical disks; carrier wave signal processing modules; and hardware devices that are specially configured to store and execute program code, such as Application-Specific Integrated Circuits (ASICs), Programmable Logic Devices (PLDs), Read-Only Memory (ROM) and Random-Access Memory (RAM) devices.

(95) Examples of computer code include, but are not limited to, micro-code or micro-instructions, machine instructions, such as produced by a compiler, code used to produce a web service, and files containing higher-level instructions that are executed by a computer using an interpreter. For example, embodiments may be implemented using imperative programming languages (e.g., C, Fortran, etc.), functional programming languages (Haskell, Erlang, etc.), logical programming languages (e.g., Prolog), object-oriented programming languages (e.g., Java, C++, etc.) or other suitable programming languages and/or development tools. Additional examples of computer code include, but are not limited to, control signals, encrypted code, and compressed code.

(96) While various specific examples and embodiments of systems and tools for selective tissue ablation with irreversible electroporation were described in the foregoing for illustrative and exemplary purposes, it should be clear that a wide variety of variations and alternate embodiments could be conceived or constructed by those skilled in the art based on the teachings herein. While specific methods of control and DC voltage application from a generator capable of selective excitation of sets of electrodes were disclosed, persons skilled in the art would recognize that any of a wide variety of other control or user input methods and methods of electrode subset selection etc. can be implemented without departing from the scope of the present invention. Likewise, while the foregoing described a range of specific tools or devices for more effective and selective DC voltage application for irreversible electroporation through ionic fluid irrigation and ultrasonic agitation, including insulating balloon constructions, focal ablation tools, and a basket catheter with a multiplicity of, other device constructions or variations could be implemented by one skilled in the art by employing the principles and teachings disclosed herein without departing from the scope of the present invention, in the treatment of cardiac arrhythmias, in intravascular applications, or a variety of other medical applications.

(97) Furthermore, while the present disclosure describes specific embodiments and tools involving irrigation with saline fluids and the use of temperature to selectively ablate tissue by taking advantage of the temperature-dependence of the threshold of irreversible electroporation, it should be clear to one skilled in the art that a variety of methods and devices for steady fluid delivery, or for tissue heating through the delivery of focused kinetic energy or electromagnetic radiation could be implemented utilizing the methods and principles taught herein without departing from the scope of the present invention.

(98) Where schematics and/or embodiments described above indicate certain components arranged in certain orientations or positions, the arrangement of components may be modified. For example, although the controller **900** is shown as optionally including the pacing module **902**, in other embodiments, the controller **900** can interface with a separate pacing module. Similarly, where methods and/or events described above indicate certain events and/or procedures occurring in certain order, the ordering of certain events and/or procedures may be modified.

(99) Although various embodiments have been described as having particular features and/or combinations of components, other embodiments are possible having a combination of any features and/or components from any of embodiments as discussed above.

Claims

1. A method for conducting bipolar pulsed field ablation of cardiac tissue through a process of irreversible electroporation, the method comprising: advancing a medical device comprising a catheter defining a proximal end and a distal end, the catheter comprising a plurality of electrodes disposed at the distal end; connecting an electroporation system in electrical communication with the plurality of electrodes of the medical device via one or more leads extending within the catheter; wherein the electroporation system comprises a generator; generating, via the generator, at least one sequence of biphasic voltage pulses having at least one amplitude, the at least one amplitude between about 0.5 kV and about 10 kV; applying, via the electroporation system, the at least one sequence of biphasic voltage pulses to the plurality of electrodes such that current associated with at least one first voltage pulse from the at least one sequence of voltage pulses flows between a first anode electrode and first and second cathode electrodes of the plurality of electrodes to thereby generate an electric field of sufficient strength to cause irreversible electroporation of the cardiac tissue proximate the distal end of the catheter; applying, via the electroporation system, the at least one sequence of biphasic voltage pulses to the plurality of electrodes such that current associated with at least one second voltage pulse from the at least one sequence of biphasic voltage pulses flows between a second anode electrode and third and fourth cathode electrodes of the plurality of electrodes to thereby generate an electric field of sufficient strength to cause irreversible electroporation of the cardiac tissue proximate the distal end of the catheter, wherein the second anode electrode differs from the first anode electrode and wherein the third and fourth cathode electrodes differ from the first and second cathode electrodes.
2. The method of claim 1, wherein the at least one sequence of biphasic voltage pulses are square waves having a positive phase and a negative phase.
3. The method of claim 1, wherein the at least one amplitude is between about 1 kV and about 2 kV.
4. The method of claim 1, wherein the electroporation system comprises a user interface, and further comprising selecting, via the user interface of the electroporation system, the first anode electrode and the first and second cathode electrodes of the plurality of electrodes.
5. The method of claim 4, wherein the user interface comprises a touchscreen display.
6. A method for conducting bipolar pulsed field ablation of cardiac tissue through a process of irreversible electroporation, the method comprising: advancing a medical device comprising a catheter defining a proximal end and a distal end, the catheter comprising a plurality of electrodes disposed at the distal end; electrically coupling an electroporation system with the plurality of electrodes of the medical device via one or more leads extending within the catheter, wherein the electroporation system comprises a user interface and a generator; selecting, via the user interface of the electroporation system, a first of anode-electrode from the plurality of electrodes and a plurality of cathode electrodes from the plurality of electrodes; generating, via the generator, at least one sequence of biphasic voltage pulses having at least one amplitude, the at least one amplitude between about 0.5 kV and about 10 kV; applying, via the electroporation system, one or more voltage pulses from the first at least one sequence of biphasic voltage pulses to the first anode electrode and the plurality of cathode electrodes of the plurality of electrodes so that current associated with each of the one or more voltage pulses flows between the first anode electrode and the plurality of cathode electrodes.
7. The method of claim 6, wherein the plurality of anode-cathode electrodes includes two electrodes.
8. The method of claim 6, wherein the catheter further comprises a slidable first member and a second member carrying the plurality of electrodes, the second member having a first end attached to the distal end of the catheter and an opposite second end coupled with the slidable first member.

9. The method of claim 8, wherein movement of the slidable first member relative to the distal end of the catheter causes the second member to move between a first configuration in which the plurality of electrodes are aligned with the length of the catheter and a second configuration in which the plurality of electrodes are disposed radially outward of the catheter.
10. The method of claim 6, further comprising synchronizing the generator output with either a paced stimulus applied to a heart or to an external trigger signal.
11. The method of claim 10, wherein the one or more voltage pulses from the at least one sequence of biphasic voltage pulses are applied to the first anode electrode and the plurality of cathode electrodes during predefined intervals from the paced stimulus or the external trigger signal.
12. The method of claim 6, further comprising applying, via the electroporation system, the one or more voltage pulses from the at least one sequence of biphasic voltage pulses to the first anode electrode and the plurality of cathode electrodes while the remaining electrodes of the plurality of electrodes are inactive.
13. A method for conducting bipolar pulsed field ablation of cardiac tissue through a process of irreversible electroporation, the method comprising: positioning a medical device comprising a catheter defining a proximal end and a distal end, the catheter comprising a plurality of electrodes disposed at the distal end, the plurality of electrodes including at least a first electrode, a second electrode, and a third electrode; connecting an electroporation system in electrical communication with the plurality of electrodes of the medical device via one or more leads extending within the catheter; wherein the electroporation system comprises a generator; generating, via the generator, at least one sequence of biphasic voltage pulses having at least one amplitude, the at least one amplitude between about 0.5 kV and about 10 kV; applying, via the electroporation system, at least one first voltage pulse from the at least one sequence of biphasic voltage pulses concurrently between the first electrode as an anode electrode and the second and third electrodes as cathode electrodes to thereby generate an electric field of sufficient strength to cause irreversible electroporation of the cardiac tissue proximate the distal end of the catheter.
14. The method of claim 13, further comprising applying, via the electroporation system, at least one second voltage pulse from the at least one sequence of biphasic voltage pulses concurrently between the second electrode as an anode electrode and the first and third electrodes as cathode electrodes.
15. The method of claim 13, wherein the plurality of electrodes includes nine or more electrodes.
16. The method of claim 13, wherein the electroporation system comprises a user interface configured to display a representation of the plurality of electrodes and an indication of the electrodes to be used in ablation.
17. The method of claim 16, wherein the user interface is configured to display an impedance detected at each electrode of the plurality of electrodes.
18. The method of claim 16, further comprising selecting, via the user interface, the first, second, and third electrodes.
19. The method of claim 13, wherein the at least one first voltage pulse comprises multiple voltage pulses.
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